

Budget-Robust Station Selection for Microtransit Under Demand Uncertainty

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1 Introduction

Virtual bus stop (VBS) selection is a key design problem in microtransit systems because selected stations determine service coverage, passenger walking access, and the ability to pool demand efficiently. However, station selection is a strategic decision made before demand is fully realized, while station activation and OD assignment are operational decisions that can adapt after demand is observed. A station set that performs well under nominal demand may therefore be fragile when demand shifts across OD pairs or time periods.

This motivates a robust station selection model. Instead of optimizing only for a fixed estimated demand vector, we allow OD demand to vary within a budgeted uncertainty set. The first-stage decision selects a long-term set of candidate stations, while the second-stage decisions activate stations and assign OD demand under each realized demand pattern. This timing structure protects service coverage against plausible demand deviations while avoiding the excessive conservatism of assuming that all OD pairs simultaneously attain their worst-case values.

We compare the nominal and robust formulations to evaluate whether robustness improves out-of-sample service coverage quality. In particular, we study how the robust model affects assignment quality, high-percentile passenger cost, and the price paid for robustness.

2 Model

2.1 Nominal Model

2.1.1 Decision Variables

$y_j \in \{0, 1\}$	build station j
$z_{js} \in \{0, 1\}$	station j is active in scenario s
$x_{odjks} \in \{0, 1\}$	assign OD pair (o, d) to origin VBS j , destination VBS k in scenario s

Walking distance induces admissibility sets. Let $\Delta(o) \subseteq J$ be the candidate pickup stations within the walking limit of origin o , and let $\Delta(d) \subseteq J$ be the candidate dropoff stations within the walking limit of destination d . Define the admissible station-pair set

$$A_{od} := \{(j, k) : j \in \Delta(o), k \in \Delta(d)\}.$$

Only pairs in A_{od} may be used to serve OD pair (o, d) .

2.2 Nominal objective

The nominal two-stage objective minimizes expected passenger travel cost:

$$\min_{x,y,z} \sum_{s \in S} \sum_{(o,d) \in \Omega} \sum_{(j,k) \in A_{od}} q_{ods} \underbrace{(d_{oj}^{\text{orig}} + d_{dk}^{\text{dest}} + \lambda c_{jk})}_{a_{odjks}} x_{odjks}, \quad (1)$$

where d_{oj}^{orig} is the walking distance from origin o to VBS j , d_{dk}^{dest} is the walking distance from VBS k to destination d , c_{jk} is the in-vehicle travel time between VBS j and k , and λ is a weight that converts routing time to the same unit as walking distance.

2.3 Constraints

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s \quad (2)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s \quad (3)$$

$$z_{js} \leq y_j \quad \forall j, s \quad (4)$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \quad \forall s \quad (5)$$

2.4 Making the model robust against demand uncertainty

2.4.1 Uncertainty Set

For each OD pair (o, d) and scenario type s , let $\bar{q}_{ods}$ denote an upper bound on demand, and let $\bar{Q}_s$ denote an upper bound on *total* demand in scenario type s . The uncertainty set is

$$\mathcal{U}_s = \left\{ q_s : \underline{q}_{ods} \leq q_{ods} \leq \bar{q}_{ods} \quad \forall (o, d), \quad \sum_{(o,d) \in \Omega} q_{ods} \leq \bar{Q}_s \right\}. \quad (6)$$

This uncertainty set says two things simultaneously:

- each OD pair must remain within its own historically calibrated range;
- the aggregate demand in the scenario cannot exceed $\bar{Q}_s$.

Observed demand is integer-valued, but the robust inner maximization may still be solved as a linear program. The key point is that the uncertainty set (6) is a bounded knapsack-type polytope. If the bounds $\bar{q}_{ods}$ and $\bar{Q}_s$ are integral, the inner LP has an optimal extreme point that is integral as well. Hence dropping integrality in the inner maximization does not change the worst-case value, even though the uncertainty set is written in continuous form.

2.4.2 Robust Counterpart

The robust model replaces the nominal objective with

$$\min_{y,z} \sum_{s \in S} \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s), \quad (7)$$

subject to the first-stage constraints (4)–(5), where $\Phi_s(z_s, q_s)$ is the recourse value defined below.

2.5 Recourse value function

The key modeling point is that x is a *recourse* decision. For fixed scenario-active stations z_s and realized demand vector q_s , define the recourse value function

$$\Phi_s(z_s, q_s) := \min_x \sum_{(o,d) \in \Omega} \sum_{(j,k) \in A_{od}} q_{ods} a_{odjks} x_{odjks} \quad (8)$$

subject to

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), \quad (9)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, \quad (10)$$

with $x_{odjks} \geq 0$. Integrality is not needed here: for fixed (o, d, s) the feasible region is a simplex intersected with bound constraints, so an optimal extreme point selects a single active station pair.

For fixed z_s , this problem decomposes by OD pair. Define the induced minimum assignment cost

$$\begin{aligned} m_{ods}(z_s) &:= \min_{x_{od..s}} \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \\ \text{s.t.} \quad &\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \\ &x_{odjks} \leq z_{js} \quad \forall (j, k) \in A_{od}, \\ &x_{odjks} \leq z_{ks} \quad \forall (j, k) \in A_{od}. \end{aligned} \quad (11)$$

Because all demand coefficients are nonnegative and the recourse constraints are separable across OD pairs, the recourse value can be written as

$$\Phi_s(z_s, q_s) = \sum_{(o,d) \in \Omega} q_{ods} m_{ods}(z_s). \quad (12)$$

For fixed z_s , the inner worst-case demand problem in (7) becomes

$$\max_{q_s \in \mathcal{U}_s} \sum_{(o,d)} q_{ods} m_{ods}(z_s) \quad \text{s.t.} \quad \underline{q}_{ods} \leq q_{ods} \leq \bar{q}_{ods}, \quad \sum_{(o,d)} q_{ods} \leq \bar{Q}_s. \quad (13)$$

This inner problem will be moved into the constraints through an epigraph variable and then dualized.

2.6 Derivation of the robust counterpart

$$\min_{y, z, \eta} \sum_{s \in S} \eta_s \quad (14)$$

subject to (4)–(5) and

$$\eta_s \geq \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s) \quad \forall s. \quad (15)$$

Using (12), each (15) becomes

$$\eta_s \geq \max_r \left\{ \sum_{(o,d)} m_{ods}(z_s) r_{ods} : \underline{q}_{ods} \leq r_{ods} \leq \bar{q}_{ods} \quad \forall (o, d), \quad \sum_{(o,d)} r_{ods} \leq \bar{Q}_s \right\} \quad (16)$$

Dualizing the inner LP in (16) gives

$$\begin{aligned} \eta_s \geq & \min_{\alpha_s \geq 0, \beta_{ods} \geq 0, \gamma_{ods} \geq 0} \quad \overline{Q}_s \alpha_s + \sum_{(o,d)} \overline{q}_{ods} \beta_{ods} - \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \\ \text{s.t.} \quad & \alpha_s + \beta_{ods} - \gamma_{ods} \geq m_{ods}(z_s) \quad \forall (o, d) \end{aligned} \quad (17)$$

By strong duality, (17) is equivalent to the existence of $\alpha_s \geq 0$, $\beta_{ods} \geq 0$, and $\gamma_{ods} \geq 0$ such that

$$\eta_s \geq \overline{Q}_s \alpha_s + \sum_{(o,d)} \overline{q}_{ods} \beta_{ods} - \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \quad \forall s, \quad (18)$$

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq m_{ods}(z_s) \quad \forall (o, d), s. \quad (19)$$

To remove the value function $m_{ods}(z_s)$, introduce epigraph variables t_{ods} and assignment variables $x_{odjks} \geq 0$:

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq t_{ods} \quad \forall (o, d), s, \quad (20)$$

$$t_{ods} \geq \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \quad \forall (o, d), s, \quad (21)$$

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s, \quad (22)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s. \quad (23)$$

Because the model minimizes η_s , (18) is tight at optimality, so η can be eliminated from the objective.

Robust counterpart MILP. The resulting formulation is

$$\min_{x, y, z, \alpha, \beta, \gamma, t} \sum_s \overline{Q}_s \alpha_s + \sum_s \sum_{(o,d)} \overline{q}_{ods} \beta_{ods} - \sum_s \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \quad (24)$$

subject to

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq t_{ods} \quad \forall (o, d), s, \quad (25)$$

$$t_{ods} \geq \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \quad \forall (o, d), s, \quad (26)$$

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s, \quad (27)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s. \quad (28)$$

plus (4)–(5), with $x_{odjks}, \alpha_s, \beta_{ods}, \gamma_{ods}, t_{ods} \geq 0$.

3 Experiments

We evaluate the proposed model on operational data from a microtransit system in Zhuzhou, China. The raw dataset contains candidate station locations, a directed routing network between stations, and realized trip requests with timestamps and origin/destination stations. Because the full network is already expensive for repeated robust optimization runs, we also construct a 40-station sampled

instance together with a sparser 40-station variant. The sampled instance preserves the original OD support restricted to the retained stations, while the sparse instance further reduces effective OD coverage. This sparse instance is used for the main model comparison because it preserves most of the observed order volume while substantially shrinking the OD support that must be calibrated from limited historical data.

Table 1: Dataset variants for the April calibration window and May backtest window.

Metric	Full Zhuzhou	Sampled Zhuzhou	Sparse Zhuzhou
Stations	84	40	40
Directed segments	6,972	1,560	1,560
April orders	2,741	2,560	2,560
April ODs with orders	473	389	130
April zero-both stations	15	1	6
May orders	31,546	30,100	30,100
May ODs with orders	1,412	949	131
May zero-both stations	11	1	6

3.1 Experimental Design

Our main comparison uses the sparse 40-station instance (`zhuzhou_data_40_sparse_v3`). We optimize on the first three days of April 2025 (April 1–3) and then evaluate the resulting station plans on the full month of May 2025 (May 1–31). This split is intentionally data-poor on the calibration side: in practice, operators often do not observe dense historical demand for the full OD matrix, especially when demand is highly intermittent at the OD-pair level.

We compare a nominal model against the robust total-demand-cap model. In the sparse study, we fix $l = 30$ candidate stations, use a walking limit of 300 seconds, set the in-vehicle time weight to $\lambda = 10$, and vary the number of active stations per scenario as $k \in \{10, 15, 20\}$. For the robust model, we vary the shared calibration quantile $q \in \{0.90, 0.95, 0.99\}$ used for both the per-OD upper bounds and the scenario-level demand cap.

Table 2: Main sparse-network experiment design.

Item	Setting
Calibration window	April 1–3, 2025
Backtest window	May 1–31, 2025
Instance	40-station sparse Zhuzhou network
Model comparison	Nominal vs. robust total-demand-cap
Candidate stations	$l = 30$
Active stations	$k \in \{10, 15, 20\}$
Walking limit	300 seconds
In-vehicle weight	$\lambda = 10$
Robust calibration	$q \in \{0.90, 0.95, 0.99\}$
In-sample orders	2,560
Out-of-sample orders	30,100

3.2 Uncertainty-Set Estimation

The robust model uses four time-of-day scenarios (morning, afternoon, evening, night). For each period s and OD pair (o, d) , we aggregate historical requests into daily OD counts. The lower bounds are fixed at zero,

$$\underline{q}_{ods} = 0,$$

which matches the implemented calibration routine and reflects the fact that many OD pairs are absent on a large fraction of days. The upper bounds $\bar{q}_{ods}$ are estimated as empirical quantiles of the zero-padded daily OD counts within each period. The scenario-level cap $\bar{Q}_s$ is estimated using the same quantile applied to total period demand. In the baseline diagnostic plots below, we use the 90th percentile calibration to illustrate the shape of the uncertainty set; in the sparse comparison, we then repeat the robust runs for $q \in \{0.90, 0.95, 0.99\}$.

Table 3 summarizes the 90th-percentile calibration on the two 40-station variants. The sampled and sparse instances share the same number of stations and directed segments, but they induce very different observed OD support. This matters because the robust calibration routine imputes missing OD upper bounds after estimating quantiles from the observed April demand. As a result, both 40-station variants produce very large box-to-cap ratios, but the sparse instance starts from much thinner observed OD coverage in every period. In both variants, the aggregate cap is binding in every period, but the sparse instance reaches that regime with substantially fewer observed OD pairs.

Table 3: Uncertainty-set diagnostics for the 90th-percentile calibration on the two 40-station variants.

Period	Sampled ODs	Sparse ODs	Sampled ratio	Sparse ratio
Morning	94	80	17.42	17.46
Afternoon	169	103	7.12	7.28
Evening	249	122	2.86	2.97
Night	120	87	13.16	13.31

4 Results

We compare the nominal and robust models on the sparse 40-station instance using a single compressed summary table. For each (k, q) pair, we report the April and May mean weighted cost per order, the corresponding standard deviations, and the number of uncovered stations summed across the four scenarios. Nominal and robust results are listed on separate rows so the tradeoffs can be read directly without splitting the discussion across separate plots and subsections. In the sparse nominal solutions, these uncovered stations appear only in the lower-demand afternoon and night periods; the robust solutions eliminate them in all scenarios.

5 Conclusion

Table 4: Sparse-network nominal-versus-robust comparison. Uncovered stations are summed across the four time periods.

k	q	Model	April mean cost	May mean cost	April std.	May std.	Uncovered stations
10	–	Nominal	4561.15	4286.38	849.13	894.21	3
10	0.90	Robust	4823.04	4256.25	807.51	892.55	0
10	0.95	Robust	4832.25	4312.02	791.53	901.22	0
10	0.99	Robust	4803.70	4242.85	800.80	905.16	0
15	–	Nominal	4319.07	4241.52	853.06	907.28	1
15	0.90	Robust	4549.15	4271.61	798.78	888.68	0
15	0.95	Robust	4541.65	4272.15	764.65	887.14	0
15	0.99	Robust	4532.36	4273.29	789.63	886.12	0
20	–	Nominal	4261.51	4241.34	883.78	907.12	0
20	0.90	Robust	4347.70	4241.16	811.70	907.61	0
20	0.95	Robust	4449.54	4243.56	758.56	903.82	0
20	0.99	Robust	4394.34	4243.84	774.26	902.89	0

Table 5: Dense sampled-network nominal-versus-robust comparison for the 2026-05-02 experiment. Uncovered stations are summed across the four time periods.

k	q	Model	April mean cost	May mean cost	April std.	May std.	Uncovered stations
10	–	Nominal	4331.02	4087.99	1150.89	1083.71	0
10	0.90	Robust	4477.52	4207.09	1061.26	1089.36	0
10	0.95	Robust	4568.88	4083.38	1064.82	1082.61	0
10	0.99	Robust	4536.73	4220.16	1050.22	1108.41	0
15	–	Nominal	4117.31	4095.45	1105.11	1075.04	0
15	0.90	Robust	4393.37	4080.44	1022.25	1086.52	0
15	0.95	Robust	4294.70	4085.23	1028.35	1081.02	0
15	0.99	Robust	4275.41	4084.24	1017.72	1082.60	0
20	–	Nominal	4056.09	4079.58	1122.79	1086.53	0
20	0.90	Robust	4174.13	4084.27	1015.06	1081.13	0
20	0.95	Robust	4257.75	4094.05	1002.38	1065.27	0
20	0.99	Robust	4255.09	4081.41	1006.12	1085.00	0